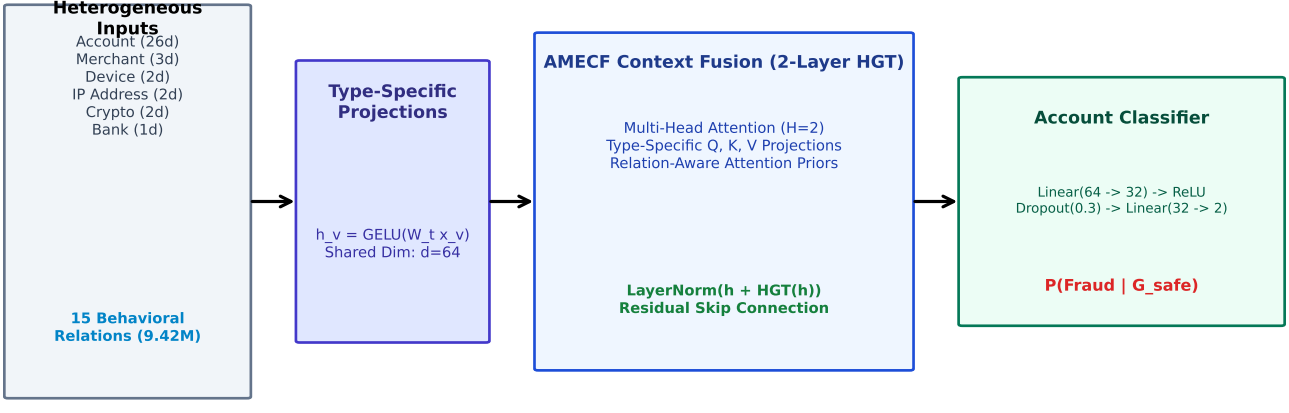
Heterogeneous Inputs
Account (26d)
Merchant (3d)
Device (2d)
IP Address (2d)
Crypto (2d)
Bank (1d)
15 Behavioral Relations (9.42M)
Type-Specific Projections
h_v = GELU(W_t x_v)
Shared Dim: d=64
AMECF Context Fusion (2-Layer HGT)
Multi-Head Attention (H=2)
Type-Specific Q, K, V Projections
Relation-Aware Attention Priors
LayerNorm(h + HGT(h))
Residual Skip Connection
Account Classifier
Linear(64 -> 32) -> ReLU
Dropout(0.3) -> Linear(32 -> 2)
P(Fraud | G_safe)